

# User Guide

**For PPG Signal Analysis Monitor**

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# PPG Live Plot - Monitor User Guide

**Version:** 1.10

**Scope:** the Python front end in Monitor/python\_plot/, and how to run it.

**Audience:** Anyone who needs to watch a PPG recording being analysed, without reading the engine.

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This guide covers running the system. How it works is covered in Technical Document.

## 1. Introduction

**The engine** is ppg\_analysis, the compiled C program that does all the real work: it reads a PPG recording, finds every heartbeat, and works out heart rate, respiratory rate and their variability. On its own it writes its results to files and prints a log. Wherever this guide refers to **the engine**, this is the program it is referring to.

The Python front end adds an additional feature: **a live window** that draws the waveform as it is analysed, marks every systolic peak and foot on it, and shows the current measurements.

Those measurements appear on **cards** - the rounded boxes at the top of the window, one for each number, each showing a title and its current value. Section 5 shows them in a picture.

Python does not do any signal processing. It starts the C program, reads what it produces, and draws it.

| Question                        | Answer                                                     |
|---------------------------------|------------------------------------------------------------|
| Does Python change the numbers? | No. It only draws them.                                    |
| Can I run without Python?       | Yes - see section 7. The C program is complete on its own. |
| Does the C program need Python? | No. It contains no reference to it.                        |

## 2. Requirements

| Requirement                      | Check it with                |
|----------------------------------|------------------------------|
| Python 3.10 or newer             | python --version             |
| PySide6                          | python -c "import PySide6"   |
| pyqtgraph                        | python -c "import pyqtgraph" |
| MinGW-w64 (gcc and mingw32-make) | gcc --version                |

If PySide6 and pyqtgraph are missing:

```
pip install PySide6 pyqtgraph
```

**Installing MinGW-w64.** This is the C compiler and the make tool, used to build the engine. On Windows the simplest way to get both is winget, from a PowerShell prompt:

```
winget install BrechtSanders.WinLibs.POSIX.UCRT
```

The project was developed and tested using this build environment. It automatically adds GCC and MinGW32-Make to the system PATH. After installation, a new terminal session must be opened for the changes to take effect. The installation can then be verified by checking that both GCC and MinGW32-Make are available through the terminal.

```
gcc --version
mingw32-make --version
```

There are two important points regarding the tool names that may cause confusion. MinGW uses mingw32-make as its build utility, whereas Linux and MSYS2 use make. Therefore, all build commands in this documentation use mingw32-make.

Additionally, the CC=gcc option appended to the build commands is required because the Makefile defaults to a compiler named cc, which is generally not provided by a standard MinGW installation.

It is not necessary to build the C engine manually. The front-end application automatically builds the C engine during the first execution and rebuilds it whenever any of the source files have been modified.

### 3. Execution

One command. Every command in this guide is run **from the project directory**, the one containing the Makefile, and <recording> stands for whichever recording you are analysing.

```
cd <the project directory>
python Monitor\python_plot\ppg_plot.py -i <recording> -s adult
```

The front end itself works from any directory - only the short relative paths used here need you to be in the project. If it is more convenient to keep the paths in variables:

```
$PLOT = "Monitor\python_plot\ppg_plot.py"
$REC = "<recording>"

python $PLOT -i $REC -s adult
```

**Two recordings come with it**, in Monitor/, so you have something to run straight away:

| Recording                      | Length          | Run it with |
|--------------------------------|-----------------|-------------|
| synthesized_sample_adult.txt   | About 8 minutes | -s adult    |
| synthesized_sample_neonate.txt | About 1 minute  | -s neonate  |

```
python Monitor\python_plot\ppg_plot.py -i Monitor\synthesized_sample_neonate.txt -s neonate
```

Start with the neonate one; it finishes in about a minute. Match -s to the file, because the patient type is the setting that most affects the result.

A word of warning about that file: it runs at about 197 bpm, which is above the 90-181 band the engine expects for a neonate, so it prints a warning, and the respiratory and variability cards stay at --. That is the engine behaving correctly on an unusual recording, not a fault in the tooling.

#### What happens, in order:

| Step | What you see                                                                 |
|------|------------------------------------------------------------------------------|
| 1    | Building the engine - only if it is missing or out of date (explained below) |

|   |                                                                                                                |
|---|----------------------------------------------------------------------------------------------------------------|
| 2 | ppg_plot: followed by the command it is about to run — Python has started the engine and is reading its output |
| 3 | Writing CSV files to ...run(folder)                                                                            |
| 4 | The C program's own log scrolls in the terminal                                                                |
| 5 | The plot window opens and starts filling                                                                       |

**How it decides the engine is out of date.** It compares the binary's timestamp against every C source file, every header, and the Makefile. If any of them has been edited more recently than the binary was built, it rebuilds before running. If none has, it skips the build and starts straight away. --no-build stops it doing this at all.

Close the window to stop. The recording is analysed in real time, so a recording of eight minutes takes eight minutes to watch. Add --turbo to run flat out.

There is also a make route, from the project directory:

```
mingw32-make plot CC=gcc REC=<recording>
mingw32-make plot CC=gcc REC=<recording> ARGS="-s neonate --turbo"
```

## 4. Options

**Full list at any time with -h.**

### Mandatory

Exactly one option must be supplied.

| Option      | Value       | Meaning                                                                                                                                                 |
|-------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| -i, --input | a file path | The recording to analyse. It is read, never written to. - or stdin makes the engine read standard input instead, which then needs something to feed it. |

### Optional

Everything below has a working default, so a run needs none of it.

#### Choosing what to analyse

| Option         | Default       | Meaning                                                                                                   |
|----------------|---------------|-----------------------------------------------------------------------------------------------------------|
| -s, --subject  | adult         | Patient type: neonate, child or adult.                                                                    |
| -r, --rate     | 125           | Sampling rate of the recording, in Hz.                                                                    |
| -c, --count    | 0             | How many samples to analyse; 0 means the whole file.                                                      |
| -d, --detector | terma         | Force a beat detector: ims or terma. Only needed to override the patient type's own choice.               |
| -nu, --scale   | Automatically | Input scale. Normally leave this off; the engine decides it from the recording and reports what it chose. |

**-s** deserves a second look even though it is optional. It selects the expected heart and respiratory rate bands, the analysis window and the beat detector, and it is the single setting that most affects the result. Analysing a neonate with the adult band is the largest source of respiratory-rate error there is.

#### Where the results go

| Option       | Default             | Meaning                                                                              |
|--------------|---------------------|--------------------------------------------------------------------------------------|
| -o, --prefix | --                  | Prefix for the result files, so successive runs do not overwrite each other.         |
| --outdir     | A folder named run/ | Where the result files are written. Created inside the project if it does not exist. |

#### How it runs

| Option       | Default             | Meaning                                                                          |
|--------------|---------------------|----------------------------------------------------------------------------------|
| --turbo      | off                 | Do not run in real time; analyse as fast as possible.                            |
| --no-display | off                 | No window: print a summary and exit.                                             |
| --no-stack   | off                 | Draw both waveforms at their true amplitudes instead of stacking them.           |
| --no-cards   | off                 | Ignore the measurements, so the waveforms still draw but every card stays at --. |
| --exe        | found automatically | Use a particular ppg_analysis binary.                                            |
| --no-build   | off                 | Never rebuild the engine, even if it is out of date.                             |
| -h, --help   | -                   | Print the option list and exit.                                                  |

#### Leave -nu alone

-nu multiplies every sample as it is read. It exists because a recording is plain text with no declared scale, and **getting it wrong is the quickest way to get a run with no beats in it.**

The user is no longer required to specify the scale. If the scale information is omitted, the engine automatically determines the appropriate scale from the textual content of the recording and reports the scale selected for processing.

Input scale: -nu 1000000 (auto: decimal text, 5 fractional digits, max |v| = 0.6217 over 1024 samples)

Pass -nu only if you have a specific reason to override that.

#### Choosing the patient type

-s is the setting that most affects the result. It selects the expected heart and respiratory rate bands, the analysis window, and which beat detector runs.

| -s      | Respiratory band | Heart-rate band | Detector |
|---------|------------------|-----------------|----------|
| Neonate | 22-66 /min       | 90-181 bpm      | IMS      |
| Child   | 11-53 /min       | 43-156 bpm      | TERMA    |

|       |           |            |       |
|-------|-----------|------------|-------|
| Adult | 4-30 /min | 43-104 bpm | TERMA |
|-------|-----------|------------|-------|

Analysing a neonate with the adult band is the single largest source of respiratory-rate error, so it is worth getting right.

## 5. Monitor Display



The live window, with each part numbered

| #  | Fields explanation                                                                                                                                                                                                                                                                           |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>Status pill.</b> Green and reading LIVE while the run is in progress. It turns red and reads ERROR if the engine stops unexpectedly.                                                                                                                                                      |
| 2  | <b>A card.</b> One of the eight rounded boxes at the top, each showing a title and its current value.                                                                                                                                                                                        |
| 3  | <b>A card's value.</b> Some values carry a small grey note beside them: n=87 on the HRV cards says how many intervals the figure rests on, and a reading being held because no new one arrived shows its age instead, as - 24s. After a minute with nothing fresh the card falls back to --. |
| 4  | <b>Raw PPG,</b> the grey trace, drawn on top: the recording exactly as it arrived.                                                                                                                                                                                                           |
| 5  | <b>Detection signal,</b> the green trace, drawn underneath: what the beat detectors were actually given, after filtering and smoothing.                                                                                                                                                      |
| 6  | <b>Systolic peak,</b> a red triangle, on the green trace.                                                                                                                                                                                                                                    |
| 7  | <b>Systolic foot,</b> a purple triangle, on the green trace.                                                                                                                                                                                                                                 |
| 8  | <b>Legend,</b> naming the two waveforms and the two markers.                                                                                                                                                                                                                                 |
| 9  | <b>Status line,</b> showing how far through the recording the run has got.                                                                                                                                                                                                                   |
| 10 | <b>RR Status,</b> saying in words what the engine did with the most recent window rather than showing a number.                                                                                                                                                                              |

**The two waveforms.** They scroll right to left, about ten seconds at a time.

| Waveforms        | Colour            | What it is                                                               |
|------------------|-------------------|--------------------------------------------------------------------------|
| Raw PPG          | Grey, on top      | The recording exactly as it arrived                                      |
| Detection signal | Green, underneath | What the beat detectors were given, after the band-pass and the smoother |

**The eight cards.** The current measurements, in two rows of four.

The top row is heart rate variability - four ways of describing how much the time between beats changes:

| Card        | Unit | What it is                                                             |
|-------------|------|------------------------------------------------------------------------|
| HRV Mean NN | ms   | The average time between heartbeats                                    |
| HRV SDNN    | ms   | How much those times vary overall                                      |
| HRV RMSSD   | ms   | How much they change from one beat to the next                         |
| HRV pNN50   | %    | The share of consecutive beats whose timing differs by more than 50 ms |

The second row is the vital signs:

| Card             | Unit | What it is                                             |
|------------------|------|--------------------------------------------------------|
| Heart Rate       | bpm  | How fast the heart is beating                          |
| Respiratory Rate | /min | How fast the subject is breathing                      |
| RR Variability   | ms   | How much the breathing intervals vary                  |
| RR Status        | -    | What the engine did with the latest window - see below |

**The RR Status card is worth knowing about.** It does not show a number; it says in words what the engine did with the most recent window:

| What the card shows      | What it means                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Warming up 16 of 66 s    | The analysis window has not grown to full length yet. The numbers are seconds: how much signal it has, out of what it needs. |
| Warming up               | Still no rate to report, and no progress figure available - usually the very first window.                                   |
| Strong match             | All three of the engine's independent breathing estimates agreed. The most confidence available.                             |
| Moderate match           | Two of the three agreed and were used; the third was set aside. A normal, usable reading.                                    |
| Strong match - warm-up   | As above, but measured on a window that is still growing, so treat it as provisional.                                        |
| Moderate match - warm-up | The same caveat, on a two-of-three agreement.                                                                                |



|                               |                                                                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Searching - BW differs        | The estimates did not agree closely enough to report a rate, and this names the one that fell outside the band - AM, BW or FM, or a pair such as AM/FM. |
| Searching - all three differ  | No two estimates agreed with each other at all.                                                                                                         |
| Searching - BW/FM has no peak | Fewer than two estimates had a clear enough spectral peak to enter the vote. Different from disagreeing: these never voted.                             |
| Searching                     | The estimates did not agree, and the reason could not be worked out from the window.                                                                    |
| No clear peak                 | No credible breathing rhythm was found in that window at all.                                                                                           |

A Searching or No clear peak reading is **not a fault**. It means the engine would rather report nothing than a rate it cannot stand behind. The Respiratory Rate card shows -- for those windows and fills again when the signal supports it.

### Why a card reads "--"

A card showing '--' is telling you something true. It means nothing yet supports a number, and it is deliberate that none is invented to fill the gap. -- is the only thing the window ever shows in that situation; there is no other wording and no error is raised.

| Reason                                            | When it clears                                                                                           |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Not enough of the recording has been analysed yet | A heart rate needs two beats; a respiratory rate needs a full analysis window, about 8 seconds of signal |
| The measurement was not reliable enough to report | The next window, if the signal supports it. The rr status card says which of these it is                 |

The second case is worth knowing about. The engine would rather report nothing than report a rate it cannot stand behind, so when its estimate is not trustworthy it says so instead of guessing. Roughly one adult window in ten does this, and the card simply stays at -- for that window. It is normal, and not a sign that anything has gone wrong.

## 6. Executing the C engine alone

The C program is complete on its own and needs nothing from Python.

```
cd <project>
mingw32-make CC=gcc
mkdir run
cd run
..\ppg_analysis.exe -i <recording> -s adult -c 0 -o out
```

That writes the same five CSV files and prints the same log. To capture the live data stream without a display, add -p on and redirect:

```
..\ppg_analysis.exe -i <recording> -s adult -c 0 -p on -o out > stream.txt
```

The stream can be fed to any tool. A live heart-rate readout with awk, no Python involved:

```
./ppg_analysis -i <recording> -s adult -c 0 -p on 2>/dev/null \
| awk -F, '/^R,/ {printf "t=%7.1fs HR=%3s bpm SDNN=%6s ms\n", $2, $17, $19}'
```

## 7. Debugging hints

| Observed Result                               | Description                                                        | Recommended Procedure                                                                  |
|-----------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| No beats detected                             | Nothing downstream has any input                                   | Check -r matches the recording's real rate, and that the file really is a PPG waveform |
| Every card stays at --, waveform fine         | The run is too short, or no window produced a reliable measurement | Let it run longer; check -s matches the patient                                        |
| The engine was BLOCKED, not missing           | Windows Smart App Control refused a freshly built binary           | Wait a few minutes and run the same command again                                      |
| The engine did not build                      | A compiler error                                                   | The message carries the compiler's own output                                          |
| No such recording                             | The path is wrong                                                  | Check the path; quote it if it has spaces                                              |
| Ignoring malformed plot row                   | A corrupt line on the stream                                       | Rare; the count is reported at the end. Report it if it is not zero                    |
| Markers sitting beside the peaks, not on them | Should not happen                                                  | Report it - the trace and the marks have come out of step                              |
| Window opens but stays empty                  | The engine produced no rows                                        | Read the terminal; the engine's own log says why                                       |

### Reading the engine's log

The terminal output comes from the C program, not from Python. Two lines are worth finding on every run:

```
Input scale: -nu 1000000 (auto: decimal text, 5 fractional digits, ...)
trace: 60001 rows written for 60001 samples read; 713 peaks and 713 feet marked
```

The first output indicates how the recording was scaled. The second output specifies the amount of data read and the number of beats detected. If the detected peak count is zero or unusually low, the subsequent processing results cannot be considered valid. In such cases, the issue originates upstream in the processing pipeline and is unrelated to the Python implementation.

## 8. Every command, in one place

All run from the project directory. <recording> is the file you are analysing; <prefix> is any name you want the output files to start with.

### Building

| Purpose                        | Command                    |
|--------------------------------|----------------------------|
| Build the engine               | mingw32-make CC=gcc        |
| Build with every warning fatal | mingw32-make strict CC=gcc |
| Delete the build products      | mingw32-make clean         |
| What the Makefile offers       | mingw32-make help          |

On a machine whose compiler is called cc, drop CC=gcc. The Python front end builds the engine for you, so these are only needed if you want to build alone.

### Running with the live plot

| Purpose                        | Command                                                                             |
|--------------------------------|-------------------------------------------------------------------------------------|
| The normal run                 | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult                      |
| The same, through make         | mingw32-make plot CC=gcc REC=<recording>                                            |
| Pass options through make      | mingw32-make plot CC=gcc REC=<recording> ARGS="-s neonate --turbo"                  |
| A neonate, IMS detector        | python Monitor\python_plot\ppg_plot.py -i <recording> -s neonate -d ims             |
| First 20000 samples only       | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult -c 20000             |
| Keep runs apart                | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult -o <prefix>          |
| As fast as possible, no window | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult --turbo --no-display |
| Waveforms only, no cards       | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult --no-cards           |
| True amplitudes, not stacked   | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult --no-stack           |
| Choose where the csvs go       | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult --outdir <folder>    |
| Force the input scale          | python Monitor\python_plot\ppg_plot.py -i <recording> -s adult -nu 10000            |
| Use a particular binary        | python Monitor\python_plot\ppg_plot.py -i <recording> --exe <binary>                |
| Never rebuild                  | python Monitor\python_plot\ppg_plot.py -i <recording> --no-build                    |
| List every option              | python Monitor\python_plot\ppg_plot.py -h                                           |

### Running the engine alone

| Purpose                       | Command                                                                         |
|-------------------------------|---------------------------------------------------------------------------------|
| Analyse a recording           | .\ppg_analysis.exe -i <recording> -s adult -c 0 -o <prefix>                     |
| The whole file                | add -c 0; a number analyses only that many samples                              |
| Capture the live stream       | .\ppg_analysis.exe -i <recording> -s adult -c 0 -p on -o <prefix> > stream.txt  |
| Read from a pipe, powershell  | Get-Content <recording>   .\ppg_analysis.exe -i stdin -s adult -c 0 -o <prefix> |
| Read from a pipe, cmd or bash | .\ppg_analysis.exe -i stdin -s adult -c 0 -o <prefix> < <recording>             |
| Print the version             | .\ppg_analysis.exe -v                                                           |
| List every option             | .\ppg_analysis.exe -h                                                           |

The engine writes its files into the **current directory**, so run it from a scratch folder such as run\ rather than from the project root.

PowerShell has no '<' redirect, which is why the pipe form differs there. It's Get-Content adds a byte-order mark, which the engine skips.

## Checking the output

| Purpose                          | Command                                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Live heart rate from the stream  | <code>./ppg_analysis.exe -i &lt;recording&gt; -s adult -c 0 -p on 2&gt;\$null   Select-String "^R,"   ForEach-Object { (\$_ -split ",")[16] }</code> |
| The same with awk, in bash       | <code>./ppg_analysis -i &lt;recording&gt; -s adult -c 0 -p on 2&gt;/dev/null   awk -F, '/^R,/ {print \$17}'</code>                                   |
| Sample rows match the trace file | see below                                                                                                                                            |
| Metrics rows match the RR file   | see below                                                                                                                                            |

The following two checks verify that the data stream received is exactly identical to the contents of the corresponding files. These checks can be performed in a Bash terminal as follows:

```
./ppg_analysis -i <recording> -s adult -c 0 -p on -o chk > plot.txt

grep -v '^R,' plot.txt > samples.txt
awk -F, 'NR>2 { gsub(/ /, ""); print $1, "$4", "($5!=0?1:0)", "($6!=0?1:0) }' \
  chk_ppg_analysis.csv > check.txt
cmp samples.txt check.txt

grep '^R,' plot.txt | sed 's/^R,/' > metrics.txt
tail -n +2 chk_RR_Data.csv | tr -d '\r' > rrfile.txt
cmp metrics.txt rrfile.txt
```

Both cmp calls must print nothing.